Supplemental Material for "Interband Pairing Signatures in the Superfluid Response of Magic-Angle Twisted Bilayer Graphene"

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I. SCOPE OF THE SUPPLEMENTAL MATERIAL

This Supplemental Material documents the executable numerical workflow behind the PRB manuscript. It is organized as a reproducibility and self-audit record: the Bistritzer-MacDonald (BM) model conventions [1], the orbital-projected pairing construction, the finite-band response formula, the figure and table provenance, and the historical PRB benchmark reconstruction are recorded separately.

The main text makes a conservative mechanism claim: an orbital-projected interband-pairing direction produces a reproducible, normalization-conditioned change in the normalized finite-band superfluid response. The present Supplemental Material does not promote the raw response units to final experimental stiffness values. This choice is motivated by the distinction between robust quantum-geometric mechanism diagnostics [2–5] and direct quantitative comparison with recent stiffness experiments [6, 7].

The material below is meant to be read as a set of checks attached to the main text, not as a separate enlargement of the claim. The BM parameters, reporting policy, and filling crosswalk specify what the plotted axes and tabulated labels mean. The pairing-construction and finite-response sections define the diagnostic used in the main text. The major-revision sections then document the pairing-family scan, the grid/truncation/shell robustness matrix, and the response-level valley-partner sensitivity test added after review. The provenance section identifies the data and scripts behind each figure and table. The momentum-grid section supplies the selected-grid evidence invoked after the main-text $nk = 9$ table. The benchmark-reconstruction and claim-scope sections record why absolute stiffness, device-level filling, and continuum-limit interpretations are left outside the current mechanism-paper scope.

II. BM MODEL AND NUMERICAL PARAMETERS

The normal-state calculation uses the single-valley BM continuum Hamiltonian with orbital basis

$$(tA, tB, bA, bB), \tag{1}$$

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TABLE I. Default parameters used for the mechanism scans.

quantity	value
twist angle	$\theta = 1.05°$
Dirac velocity	$v_F = 2.135$ eV Å
AA tunneling	$w_0 = 87.2$ meV
AB tunneling	$w_1 = 109.0$ meV
reciprocal cutoff	$N_{\mathrm{shell}} = 3$ for dense maps; $2, 3, 4$ in robustness audit
retained bands	$n_{\mathrm{keep}} = 6$ for dense maps; $4, 6, 8$ in robustness audit
momentum grid	$nk = 7$ for dense maps; $nk = 9, 11, 13, 15$ for major-revision checks
gap scale	$\Delta_0 = 1$ meV
velocity finite difference	$dk = 10^{-6}\ \text{Å}^{-1}$

where t, b label layers and A, B label sublattices. Unless explicitly stated otherwise, the production mechanism scans use the parameters in Table I.

The moire unit-cell area and reciprocal scale used in the unit audit are

$$A_M = 1.5606 \times 10^4\ \text{Å}^2, \qquad G_M = 0.054047\ \text{Å}^{-1}. \tag{2}$$

The raw response conversion used in the audit is

$$D[\text{eV Å}^2] = \frac{D_{\mathrm{raw}}[\text{meV Å}^2]}{1000}, \tag{3}$$

with all remaining spin, valley, or convention factors kept explicit rather than absorbed into a hidden prefactor.

A. Reporting Convention for Units and Filling

Table II states the reporting convention used in the present manuscript. The main figures use normalized response ratios, while absolute eV Å^2 values are restricted to the self-audit and historical benchmark reconstruction. This is the final observable policy for the current mechanism-paper scope.

The retained-band filling proxy is computed as

$$\nu_{\mathrm{proxy}} = g\left[\left\langle N_{\mathrm{keep}}^{-1} \sum_n \Theta(\mu - \varepsilon_{n\mathbf{k}}) \right\rangle_{\mathbf{k}} - \frac{1}{2}\right], \tag{4}$$

TABLE II. Reporting convention for response and filling quantities.

quantity	present use	excluded interpretation
raw response	internal finite-band diagnostic	physical stiffness
$D_{\rm iso}(\eta)/D_{\rm iso}(0)-1$	claim-bearing mechanism observable	absolute stiffness
eV Å^2 per flavor	benchmark audit only	experimental comparison
$\nu_{\rm proxy}$	retained-band occupancy label	calibrated carrier density

with $g = 4$ used as a spin-valley counting factor in the proxy. For the $n_{\rm keep} = 6$ scans this is not identical to a calibrated experimental filling factor; it is used only to order the chemical-potential scans. The central flat-band crosswalk instead counts the two central BM bands as

$$\nu_{\rm flat} = g\left[\left\langle \sum_{n\in\text{central pair}} \Theta(\mu-\varepsilon_{n\mathbf{k}})\right\rangle_{\mathbf{k}} - 1\right], \tag{5}$$

which ranges from -4 to $+4$ for an empty-to-full central two-band subspace when $g = 4$. On the $nk = 7$, $N_{\rm shell} = 3$ mesh used for the dense response scan, the central-pair energy window is $[-1.051, 4.134]$ meV. The resulting crosswalk is shown in Table III. It should be read as a flat-band counting reference for the present BM calculation, not as a device-specific carrier-density calibration.

The filling-reference claim is checked by the self-audit

```
python3 scripts/audit_filling_sufficiency.py
```

which writes

```
data/processed/filling_sufficiency_audit.csv
```

Table IV summarizes the result. The crosswalk is sufficient as a BM central-flat-band counting reference for the present mechanism maps. It is not a device-level carrier-density calibration.

III. ORBITAL-PROJECTED PAIRING CONSTRUCTION

The pairing direction is specified first in the layer-sublattice orbital basis,

$$\Delta_{\rm orb}(\eta) = \Delta_0\,\mathcal{N}_\eta(M_0+\eta M_1), \tag{6}$$

TABLE III. Crosswalk between the retained-band filling proxy and central two-flat-band filling for the dense $nk = 7$, $n_{\mathrm{keep}} = 6$ scan.

μ (meV)	ν_{proxy}	ν_{flat}	occupied flat bands/flavor
-5	-1.483	-4.000	0.000
-4	-1.333	-4.000	0.000
-3	-1.061	-4.000	0.000
-2	-0.857	-4.000	0.000
-1	-0.653	-3.837	0.041
0	-0.381	-2.286	0.429
1	-0.054	-0.327	0.918
2	0.122	0.735	1.184
3	0.395	2.367	1.592
4	0.667	3.918	1.980
5	0.857	4.000	2.000

TABLE IV. Filling-sufficiency self-audit for the current mechanism-paper scope.

check	measured value	manuscript consequence
μ grid	11/11 values	matches dense response map
ν_{proxy}	-1.483 to 0.857	monotone scan label
ν_{flat}	-4.000 to 4.000	full central-pair range
flat-band window	-1.051 to 4.134 meV	bracketed by scan window
sampled fillings	0.327, 0.286, 0.367	near $0, -2, +2$ references
Overall scope	8/8 numeric gates	counting reference only

with

$$M_0 = \tau_0\sigma_0, \qquad M_1 = \tau_x\sigma_x. \tag{7}$$

The two normalization controls used in the manuscript are

$$\text{fixed Frobenius:} \qquad \|M_0 + \eta M_1\|_F = \|M_0\|_F, \tag{8}$$

$$\text{fixed } \Delta_0: \qquad \mathcal{N}_\eta = 1. \tag{9}$$

TABLE V. Raw valley-minus sewing diagnostic. The independently diagonalized valley-minus subspace at $-\mathbf{k}$ is compared with the target $U_+^*(\mathbf{k})$ on an $nk = 3$ grid. Small singular values and large alignment errors indicate that the raw valley-minus eigenvectors should not be used directly to define W_{inter}.

n_{keep}	alignment error	mean minimum singular value	minimum singular value
2	1.3895	0.0267	0.0218
4	1.3782	0.0201	0.0043
6	1.3760	0.0095	0.0016

The band-basis pairing matrix is

$$\Delta_{\text{band}}(\mathbf{k}) = U_+^\dagger(\mathbf{k})\Delta_{\text{orb}}U_-^*(-\mathbf{k}). \tag{10}$$

The working valley convention in the present diagnostic is the time-reversal sewn proxy

$$U_-(-\mathbf{k}) = U_+^*(\mathbf{k}). \tag{11}$$

The production projection is therefore

$$\Delta_{\text{band}}(\mathbf{k}) = U_+^\dagger(\mathbf{k})\Delta_{\text{orb}}U_+(\mathbf{k}). \tag{12}$$

This convention is useful for isolating an orbital-defined interband-pairing signature. It also avoids a known gauge artifact: independently diagonalized valley-minus eigenvectors are not automatically in the same time-reversal sewing convention as the valley-plus eigenvectors. Table V summarizes the raw-basis diagnostic.

This is still a diagnostic convention rather than a full independent two-valley implementation. Valley-asymmetric perturbations, strain, or valley-dependent interactions would require revisiting this sewing choice.

A. Response-Level Valley-Partner Sensitivity

The revision adds a response-level stress test of the valley-partner choice:

```
python3 scripts/run_valley_sewing_response_sensitivity.py
python3 scripts/audit_valley_sewing_response_sensitivity.py
python3 scripts/plot_valley_sewing_response_sensitivity.py
```

TABLE VI. Response-level valley-partner sensitivity. Percent ranges are over $nk = 9, 13, 15$, $N_{\mathrm{shell}} = 3, 4$, and $\mu = -4, 0, 2, 4$ meV.

partner	positive rows	response range (%)	mean (%)	max PH error
`tr_sewn`	24/24	4.37 to 13.60	9.18	0
`same_valley`	14/24	-28.41 to 46.31	3.74	3.731×10^{-2}
`time_reversed_valley`	12/24	-17.66 to 29.76	1.80	2.672×10^{-2}
`sewn_time_reversed_valley`	8/24	-17.28 to 16.68	-1.44	2.064×10^{-2}

The outputs are

`data/processed/valley_sewing_response_sensitivity.csv`

`data/processed/valley_sewing_response_sensitivity_summary.csv`

`data/processed/valley_sewing_response_sensitivity_audit.csv`

`figures/valley_sewing_response_sensitivity.png`

`figures/valley_sewing_response_sensitivity.pdf`

The test keeps the same finite-band response formula and changes only the paired-sector partner vectors used in the projection of Δ_{orb}. The scan covers $nk = 9, 13, 15$, $N_{\mathrm{shell}} = 3, 4$, $n_{\mathrm{keep}} = 6$, and $\mu = -4, 0, 2, 4$ meV. It is a sensitivity diagnostic, not a replacement for a complete two-valley response calculation.

The result is not that the response is valley-partner independent. It is the opposite, and the revised manuscript should say so plainly: the positive response is robust within the declared `tr_sewn` finite-band diagnostic convention, while alternative partner choices expose sign and PH-consistency sensitivity.

The projected interband-pairing weight is

$$W_{\mathrm{inter}} = \frac{\sum_{n \neq m} |\Delta_{nm}|^2}{\sum_{n,m} |\Delta_{nm}|^2}. \tag{13}$$

For the dense $nk = 7$, $n_{\mathrm{keep}} = 6$ scan, $W_{\mathrm{inter}} \simeq 0.108$ at $\eta = 1$.

B. Major-Revision Pairing-Family Scan

The review asked whether the result is tied to the single working direction $M_1 = \tau_x \sigma_x$. To answer this, the revision adds a first-pass scan of eight real, symmetric, Hermitian orbital

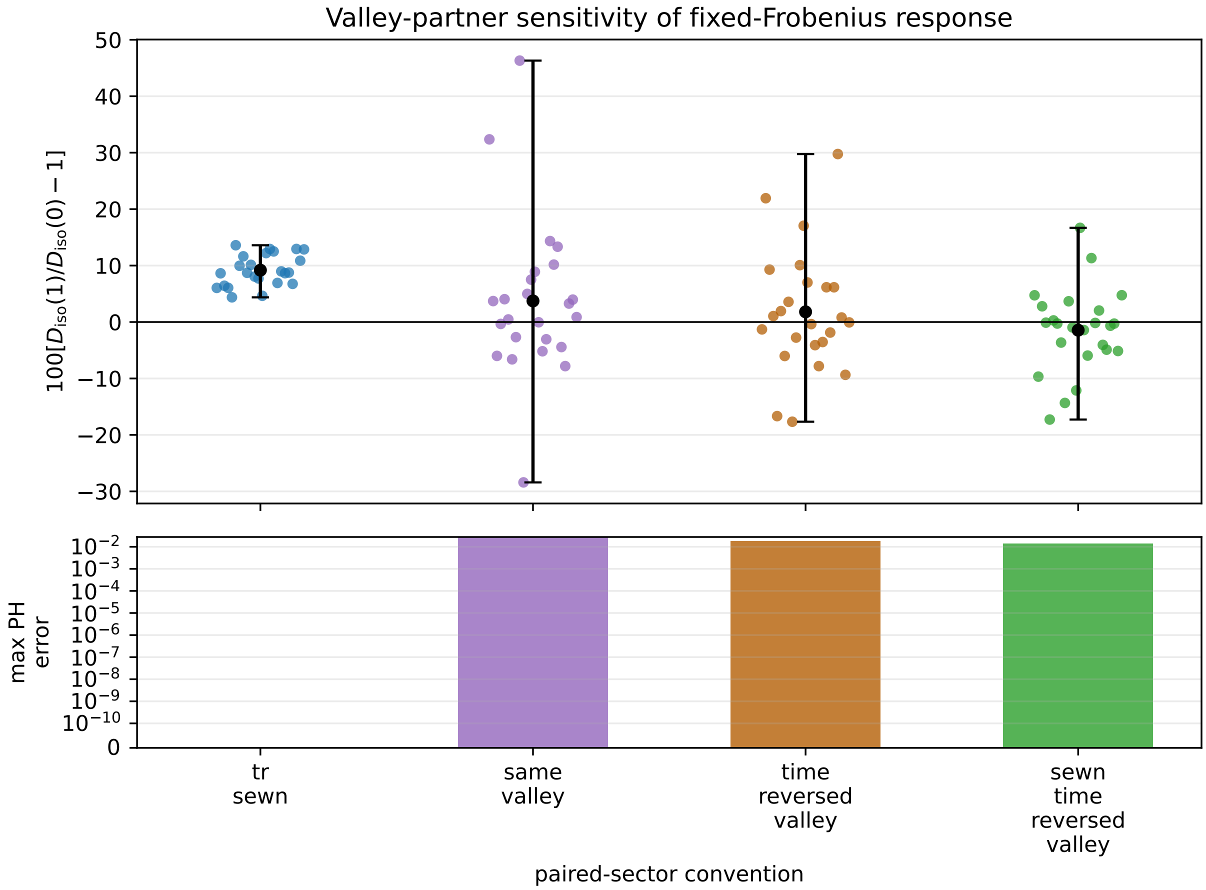

FIG. 1. Valley-partner sensitivity of the fixed-Frobenius response. The production `tr_sewn` convention remains positive and has zero particle-hole spectrum error in this diagnostic. Alternative partner choices are sign-sensitive and show measurable particle-hole spectrum errors, so they are treated as stress tests rather than replacement production conventions.

directions:

```
python3 scripts/run_pairing_family_response_scan.py
python3 scripts/audit_pairing_family_response.py
python3 scripts/plot_pairing_family_response.py
```

The outputs are

```
data/processed/pairing_family_response_scan.csv
data/processed/pairing_family_response_summary.csv
data/processed/pairing_family_response_audit.csv
```

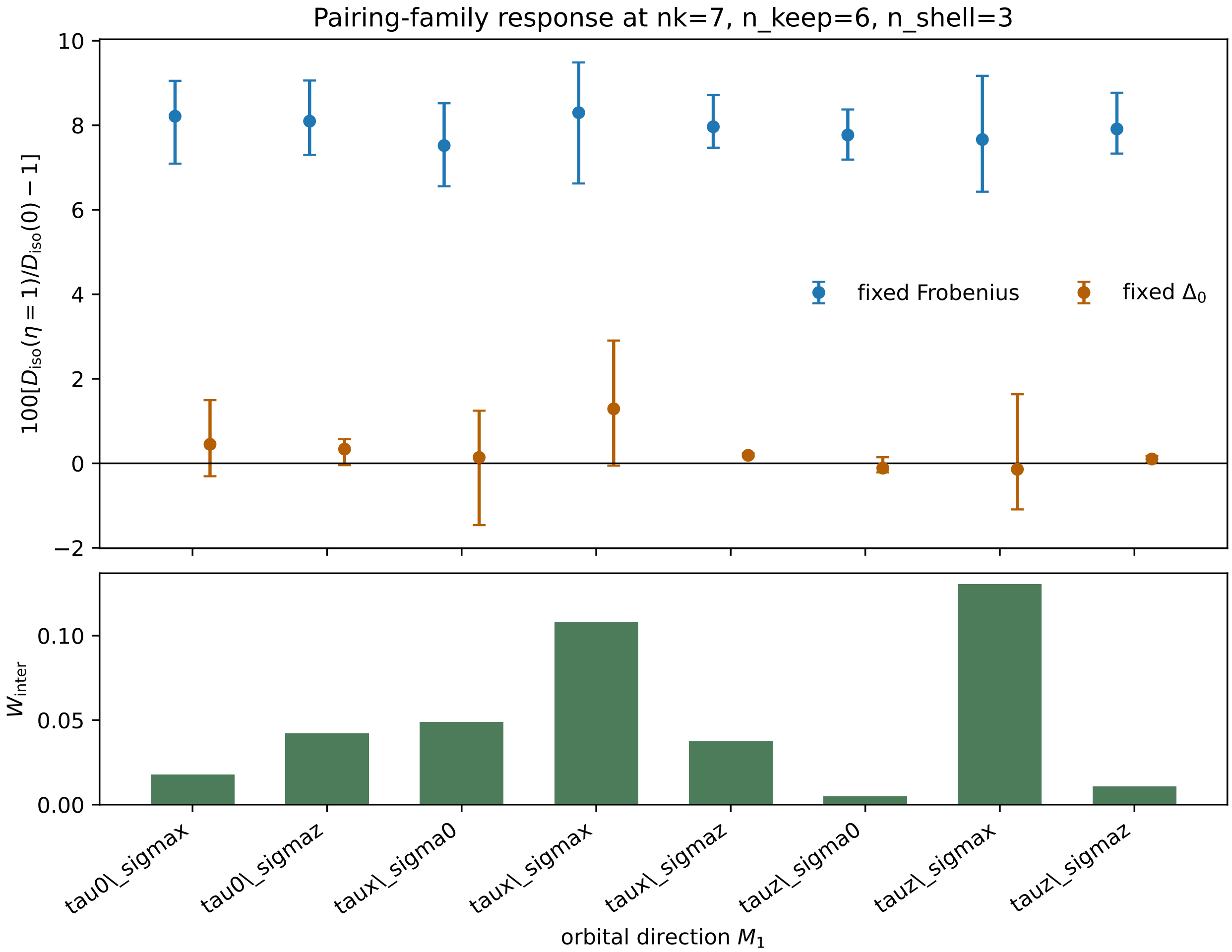

FIG. 2. First-pass pairing-family response scan. Points and bars summarize the $\eta = 1$ response relative to $\eta = 0$ across $\mu = -4, 0, 2, 4$ meV. The blue fixed-Frobenius values remain positive across all eight directions; the fixed-Δ_0 control stays close to zero and can change sign.

`figures/pairing_family_response_summary.png`

`figures/pairing_family_response_summary.pdf`

The scan uses $nk = 7$, $n_{keep} = 6$, $N_{shell} = 3$, and $\mu = -4, 0, 2, 4$ meV. It is a pairing-family test at the original dense-map resolution rather than a replacement for the grid/truncation/shell audit below. Figure 2 and Table VII show that the fixed-Frobenius response is positive for all candidate directions, while the fixed-Δ_0 control remains weak and sign-sensitive.

TABLE VII. Pairing-family response summary. Percent ranges are over $\mu = -4, 0, 2, 4$ meV.

M_1	fixed Frobenius (%)	fixed Δ_0 (%)	mean W_{inter}
$\tau_0\sigma_x$	7.09 to 9.05	-0.31 to 1.50	0.0179
$\tau_0\sigma_z$	7.30 to 9.06	-0.04 to 0.57	0.0422
$\tau_x\sigma_0$	6.56 to 8.52	-1.46 to 1.25	0.0488
$\tau_x\sigma_x$	6.62 to 9.49	-0.05 to 2.91	0.1082
$\tau_x\sigma_z$	7.47 to 8.71	0.16 to 0.23	0.0374
$\tau_z\sigma_0$	7.19 to 8.37	-0.21 to 0.14	0.0050
$\tau_z\sigma_x$	6.43 to 9.17	-1.09 to 1.63	0.1304
$\tau_z\sigma_z$	7.33 to 8.77	0.01 to 0.18	0.0109

IV. FINITE-BAND RESPONSE AND CHANNEL SPLIT

For a retained band subspace, the finite-band BdG Hamiltonian is

$$H_{\mathrm{BdG}}(\mathbf{k}) = \begin{pmatrix} \varepsilon(\mathbf{k}) - \mu & \Delta_{\mathrm{band}}(\mathbf{k}) \\ \Delta^\dagger_{\mathrm{band}}(\mathbf{k}) & -[\varepsilon(\mathbf{k}) - \mu] \end{pmatrix}. \tag{14}$$

The zero-temperature static current-current response is evaluated as

$$\Lambda_{\alpha\beta} = \sum_{a\neq b} \frac{\langle a|J_\alpha|b\rangle\langle b|J_\beta|a\rangle(f_a - f_b)}{E_b - E_a}. \tag{15}$$

The band-basis velocity is split into diagonal and off-diagonal parts,

$$v_\alpha^{\mathrm{band}} = v_\alpha^{\mathrm{intra}} + v_\alpha^{\mathrm{inter}}. \tag{16}$$

This produces conventional, geometric, and cross-channel response diagnostics. The primary normalized response map uses

$$\delta D_{\mathrm{iso}}(\mu,\eta) = \frac{D_{\mathrm{iso}}(\mu,\eta)}{D_{\mathrm{iso}}(\mu,0)} - 1, \qquad D_{\mathrm{iso}} = \frac{D_{xx} + D_{yy}}{2}. \tag{17}$$

V. FIGURE AND TABLE PROVENANCE

Table VIII summarizes the current figure and table provenance. The full machine-readable record is maintained in `notes/figure_table_provenance_2026-06-10.md`.

The dense response data are reproduced by

TABLE VIII. Manuscript artifact provenance. CSV values used in the manuscript tables are checked by `scripts/verify_manuscript_tables.py`.

artifact	data source	script	check
Frobenius heatmap	dense scan CSV	heatmap plotter	figure file exists
fixed-Δ_0 heatmap	dense scan CSV	heatmap plotter	figure file exists
dense response table	eta-summary CSV	eta summarizer	table verifier
$nk = 9$ validation table	key-point summary CSV	eta summarizer	table verifier
PRB audit table	reconstruction CSVs	audit scripts	table verifier
pairing-family revision table	pairing-family summary CSV	family scan/audit scripts	family audit
robustness-matrix revision figure	robustness summary CSV	robustness scan/audit scripts	robustness audit
valley-sensitivity revision figure	valley sensitivity summary CSV	sensitivity scan/audit scripts	sensitivity audit

```
python3 scripts/run_mu_response_scan.py \
  --nk 7 --n-shell 3 --n-keep 6 \
  --mus -5 -4 -3 -2 -1 0 1 2 3 4 5 \
  --etas 0 0.25 0.5 0.75 1 \
  --output data/processed/mu_eta_response_scan_nk7_nkeep6.csv
```

and summarized by

```
python3 scripts/summarize_eta_effect.py \
  data/processed/mu_eta_response_scan_nk7_nkeep6.csv \
  --output data/processed/mu_eta_response_scan_nk7_nkeep6_eta1_summary.csv
```

The table verifier is

```
python3 scripts/verify_manuscript_tables.py
```

VI. KEY-POINT MOMENTUM-GRID CONVERGENCE

The dense maps in the main text use $nk = 7$, while selected key points were checked at $nk = 9$, $nk = 11$, and $nk = 13$. The higher-grid scans use

```
python3 scripts/run_mu_response_scan.py \
  --nk 11 --n-shell 3 --n-keep 6 \
```

```
  --mus -4 0 2 4 --etas 0 0.5 1 \
  --output data/processed/mu_response_scan_nk11_nkeep6_keypoints.csv

python3 scripts/run_mu_response_scan.py \
  --nk 13 --n-shell 3 --n-keep 6 \
  --mus -4 0 2 4 --etas 0 0.5 1 \
  --output data/processed/mu_response_scan_nk13_nkeep6_keypoints.csv
```

and is summarized with `scripts/summarize_eta_effect.py`. The convergence figure is generated with

```
python3 scripts/plot_nk_convergence_keypoints.py
```

This section supports the main-text validation paragraph that follows the $nk = 9$ table. Its role is to test sign and scale stability at selected chemical potentials, not to replace the dense μ-η map or to turn the selected grids into a continuum extrapolation.

Figure 3 and Table IX compare the normalized $\eta = 1$ response relative to $\eta = 0$. The fixed-Frobenius response remains positive at all four key chemical potentials. The fixed-Δ_0 response remains small and sign-sensitive, consistent with the main-text interpretation that the interband-pairing signal is normalization-conditioned.

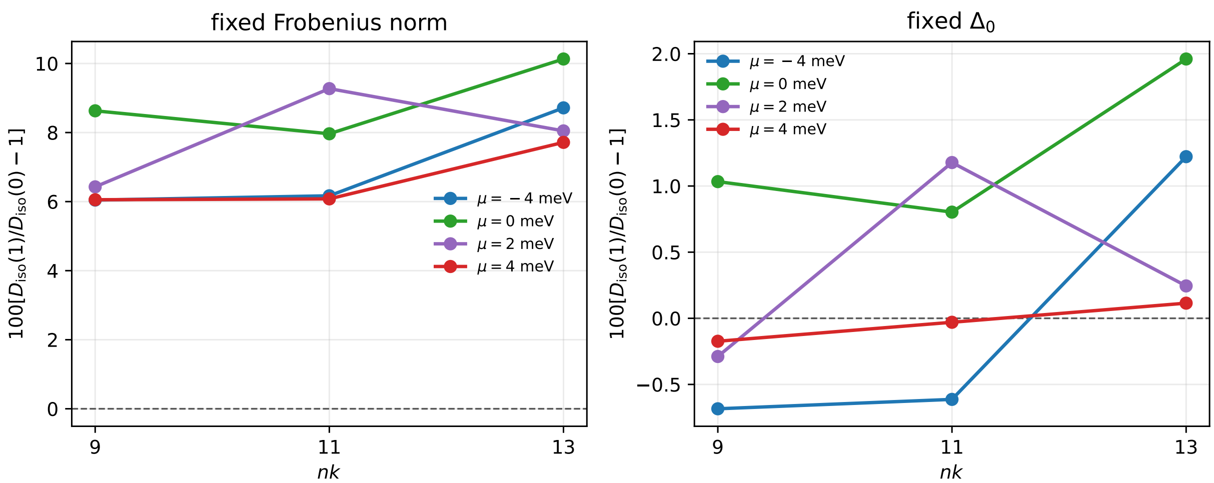

FIG. 3. Selected-grid convergence of the normalized $\eta = 1$ response. The left panel shows the fixed-Frobenius comparison used for the main mechanism claim; the right panel shows the fixed-Δ_0 control. The plotted values are drawn from the same summary data tabulated in Table IX.

TABLE IX. Key-point momentum-grid check for $D_{\rm iso}(\eta = 1)/D_{\rm iso}(\eta = 0) - 1$.

mode, μ (meV)	$nk = 9$	$nk = 11$	$nk = 13$	$13 - 9$
fixed Δ_0, -4	-0.0068	-0.0061	+0.0122	+0.0191
fixed Δ_0, 0	+0.0103	+0.0080	+0.0196	+0.0093
fixed Δ_0, 2	-0.0029	+0.0118	+0.0024	+0.0053
fixed Δ_0, 4	-0.0017	-0.0003	+0.0011	+0.0029
fixed Frobenius, -4	+0.0604	+0.0617	+0.0872	+0.0267
fixed Frobenius, 0	+0.0863	+0.0796	+0.1013	+0.0150
fixed Frobenius, 2	+0.0643	+0.0927	+0.0805	+0.0162
fixed Frobenius, 4	+0.0605	+0.0608	+0.0772	+0.0166

Across the three key-point grids the fixed-Frobenius response remains positive, with the largest $nk = 13 - nk = 9$ shift about 2.67 percentage points. Therefore the current manuscript claims sign and order-of-magnitude stability of the normalized mechanism signal, not a strict continuum-limit extrapolation of the tabulated values.

As an additional finite-grid audit, the same values are fitted to

$$y(nk) = y_0 + a/nk^2, \qquad y = 100\,[D_{\rm iso}(\eta = 1)/D_{\rm iso}(\eta = 0) - 1]\,. \tag{18}$$

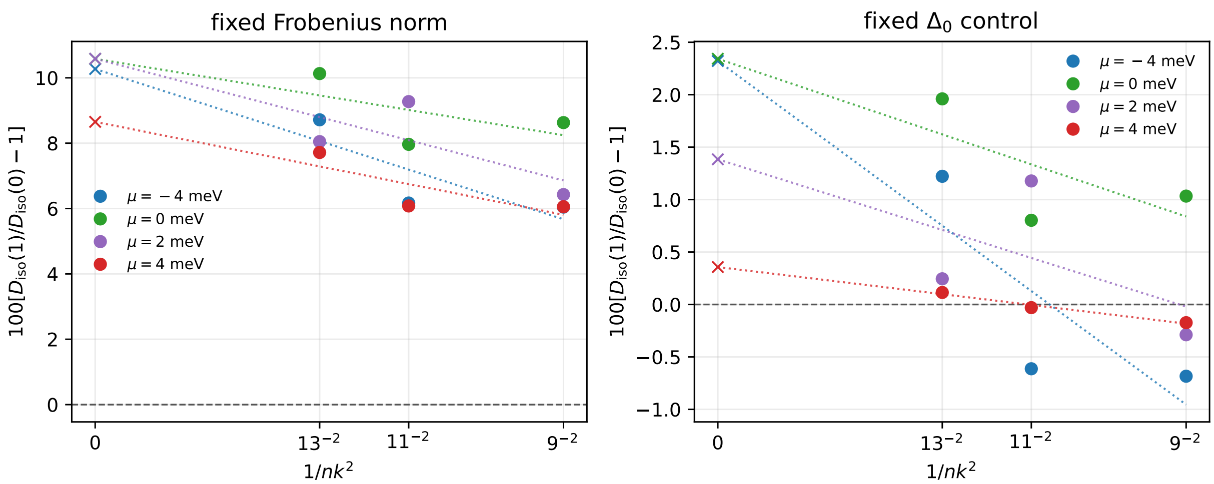

FIG. 4. Finite-grid trend audit for the selected $nk = 9, 11, 13$ key points. Markers at nonzero $1/nk^2$ are computed values, dotted lines are linear $1/nk^2$ fits, and crosses at zero are fitted intercepts. The fixed-Frobenius trend remains positive for all four chemical potentials; the fixed-Δ_0 control remains much smaller and sign-sensitive on the measured grids.

TABLE X. Finite-grid trend audit for the selected key points. Values are in percent. The $1/nk^2$ intercept is the plotted trend marker, while the $1/nk$ intercept is a sensitivity check.

mode, μ (meV)	min	max	$1/nk^2$ int.	$1/nk$ int.	sign
fixed Δ_0, -4	-0.68	1.22	2.32	4.81	mixed
fixed Δ_0, 0	0.80	1.96	2.34	3.53	positive
fixed Δ_0, 2	-0.29	1.18	1.38	2.20	mixed
fixed Δ_0, 4	-0.17	0.11	0.36	0.74	mixed
fixed Frobenius, -4	6.04	8.72	10.27	13.77	positive
fixed Frobenius, 0	7.96	10.13	10.58	12.48	positive
fixed Frobenius, 2	6.43	9.27	10.58	12.94	positive
fixed Frobenius, 4	6.05	7.72	8.65	10.82	positive

Figure 4 shows the measured points, the $1/nk^2$ trend lines, and the intercept markers at $1/nk^2 = 0$. Table X also lists a $1/nk$ intercept as a simple sensitivity check. With only three grids, these intercepts are an audit of trend stability rather than a final continuum extrapolation. The fixed-Frobenius intercepts remain positive in both trend coordinates,

TABLE XI. Targeted $nk = 15$ spot check for representative key points.

mode, μ (meV)	$nk = 13$	$nk = 15$	$15 - 13$	$W_{\text{inter}}(15)$
fixed Δ_0, 0	+0.0196	+0.0085	-0.0111	0.1076
fixed Δ_0, 2	+0.0024	+0.0071	+0.0046	0.1076
fixed Frobenius, 0	+0.1013	+0.0897	-0.0116	0.1076
fixed Frobenius, 2	+0.0805	+0.0865	+0.0060	0.1076

TABLE XII. Convergence-sufficiency self-audit for the current mechanism-paper scope.

check	measured value	manuscript consequence
Frob. sign	6.04 to 10.13%	sign stable
Frob. spread	2.85 pp	scale stable
Trend intercepts	8.65% / 10.82%	positive trend audits
$nk = 15$ spot check	8.97%, 8.65%; shift 1.16 pp	central spot stable
fixed-Δ_0 control	max 1.96%; mixed signs	weak/sign-sensitive
Overall scope	8/8 numeric gates	selected-grid claim only

whereas the fixed-Δ_0 measured-grid signs remain mixed except at $\mu = 0$.

A targeted higher-grid spot check was also run at $nk = 15$ for $\mu = 0$ and 2 meV, using only $\eta = 0$ and $\eta = 1$. This is not a full dense-grid replacement, but it tests whether the two central representative key points remain in the same response range on a denser mesh. Table XI shows that the fixed-Frobenius response remains positive and close to the $nk = 13$ values, while the fixed-Δ_0 control remains below one percent at both chemical potentials.

The selected-grid evidence is finally summarized by the convergence-sufficiency self-audit

```
python3 scripts/audit_convergence_sufficiency.py
```

which writes

```
data/processed/convergence_sufficiency_audit.csv
```

Table XII records the manuscript-level consequence. The audit supports the selected-grid normalized mechanism claim used in the main text. It does not constitute a dense-mesh replacement or a strict continuum-limit numerical value.

TABLE XIII. Major-revision robustness summary for the working direction $M_1 = \tau_x\sigma_x$.

scope	positive rows	response range (%)	consequence
all fixed Frobenius	172/180	-42.22 to 16.60	not globally truncation independent
$n_{\mathrm{keep}} \geq 6$	120/120	1.07 to 13.60	production/expanded truncations positive
$n_{\mathrm{keep}} \geq 6$, $N_{\mathrm{shell}} \geq 3$	80/80	4.37 to 13.60	production-scope response sizable
$n_{\mathrm{keep}} = 4$, $\mu = -4$ meV	8 nonpositive	-42.22 to -0.89	disclosed low-truncation boundary
all fixed Δ_0	113/180	-4.62 to 3.30	weak/sign-sensitive control

VII. MAJOR-REVISION GRID, TRUNCATION, AND SHELL ROBUSTNESS

The review asked for stronger numerical convergence and truncation control. The revision therefore adds a rectangular robustness matrix for the working direction:

```
python3 scripts/run_prb_major_revision_robustness.py
python3 scripts/audit_prb_major_revision_robustness.py
python3 scripts/plot_prb_major_revision_robustness.py
```

The outputs are

```
data/processed/prb_major_revision_robustness_matrix.csv
data/processed/prb_major_revision_robustness_summary.csv
data/processed/prb_major_revision_robustness_audit.csv
figures/prb_major_revision_robustness_matrix.png
figures/prb_major_revision_robustness_matrix.pdf
```

The matrix spans

$$nk = 9, 11, 13, 15, \qquad n_{\mathrm{keep}} = 4, 6, 8, \qquad N_{\mathrm{shell}} = 2, 3, 4, \tag{19}$$

with $\mu = -4, -2, 0, 2, 4$ meV and $\eta = 0, 0.5, 1$. The raw matrix has 1080 rows and the $\eta = 1$ versus $\eta = 0$ summary has 360 rows.

The machine-readable robustness audit passes all current scoped checks. It certifies production-scope sign stability; it does not certify a global all-truncation positive response. This distinction is essential for the revised PRB claim.

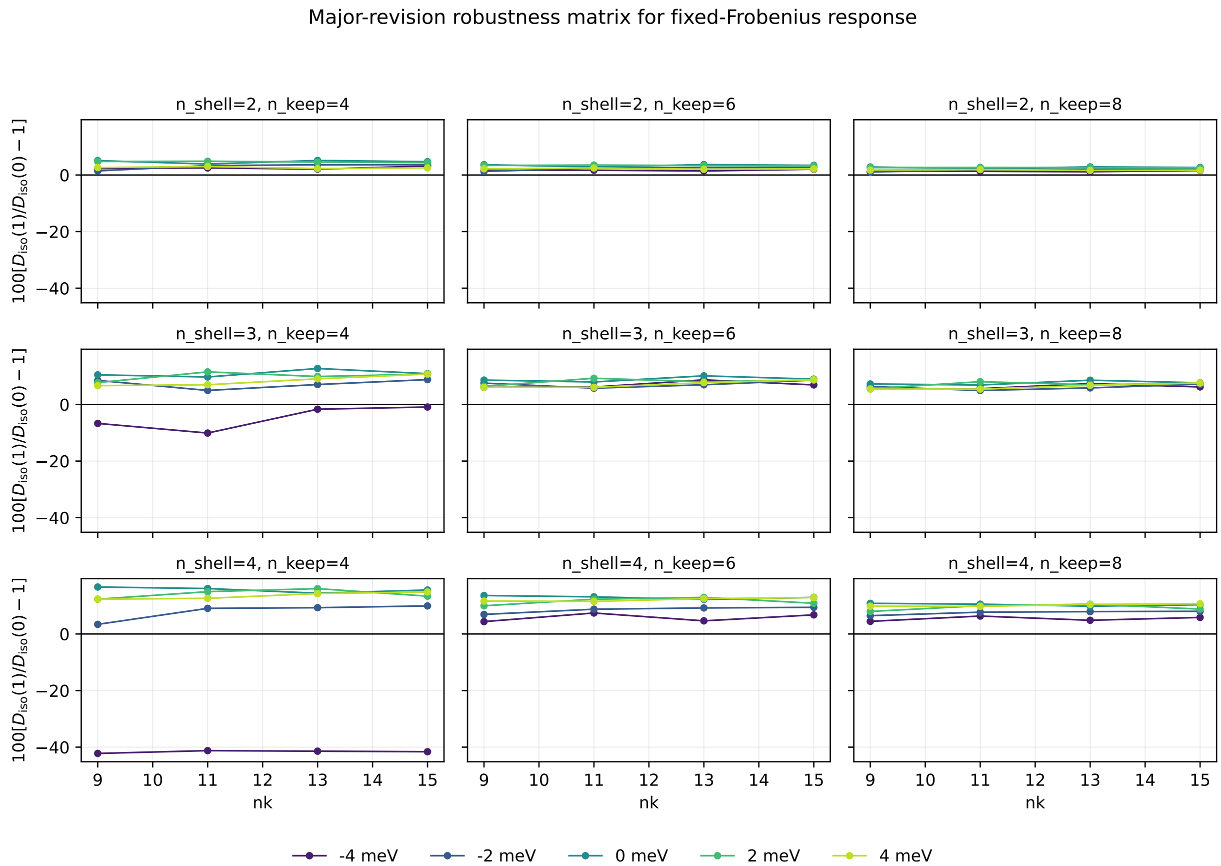

FIG. 5. Major-revision robustness matrix for the fixed-Frobenius response. Rows vary the moire shell cutoff, columns vary the retained-band truncation, and each panel shows the response versus nk at five chemical potentials. The production and expanded truncations $n_{\text{keep}} = 6, 8$ are positive throughout the tested matrix, while the low-truncation boundary appears at $n_{\text{keep}} = 4$ and $\mu = -4$ meV.

VIII. HISTORICAL PRB BENCHMARK RECONSTRUCTION

The production interband-pairing scan and the historical PRB table reconstruction are kept separate. The best simple unified reconstruction route for the old uniform-s benchmark is

$$D_{\text{conv}} = 2D_{\text{intra},\tau_z}, \qquad D_{\text{geom}} = D_{\text{inter},\tau_z}. \tag{20}$$

For $n_{\text{keep}} = 6$, this gives $D_{\text{total}} = 126.66$ eV Å^2 versus the historical target 129.30 eV Å^2, and $D_{\text{geom}} = 75.22$ eV Å^2 versus 75.30 eV Å^2.

The old $n_{\text{keep}} = 2$ endpoint is more convention-sensitive. The closest audited reconstruction combines a Gamma-centered mesh, central-pair band selection, an all-τ_z current convention, a factor of two in the intraband term, and a full curvature-like conventional correction. This gives

$$D_{\mathrm{total}} = 66.18\ \mathrm{eV}\,\mathrm{\AA}^2 \quad \text{versus} \quad 67.50\ \mathrm{eV}\,\mathrm{\AA}^2. \tag{21}$$

Because this full-curvature correction does not work uniformly at $n_{\mathrm{keep}} = 6$, the reconstruction is treated as a historical audit rather than a production convention.

IX. CLAIM SCOPE AND EXCLUDED INTERPRETATIONS

The manuscript is framed as a normalized-response mechanism paper. The text-level claim-scope audit is generated with

```
python3 scripts/audit_claim_scope.py
```

and the observable-policy audit is generated with

```
python3 scripts/audit_observable_policy.py
```

The audit outputs are

```
data/processed/claim_scope_audit.csv
data/processed/observable_policy_audit.csv
```

They are not numerical validation tools; they check that the manuscript retains the boundary statements needed to avoid overinterpreting the finite-band diagnostic calculation. Table XIV summarizes the current claim scope.

X. CURRENT LIMITATIONS AND NEXT VALIDATION GATES

The current submission-critical gaps are:

1. a full continuum-grid extrapolation beyond the present selected $nk = 9$, $nk = 11$, $nk = 13$, $nk = 15$ audits and the major-revision robustness matrix;

2. a fully independent two-valley implementation before studying valley-asymmetric perturbations or claiming valley-basis-independent stiffness physics;

TABLE XIV. Allowed manuscript claims and excluded interpretations.

topic	allowed in the present manuscript	excluded interpretation
normalized response	mechanism signature	absolute stiffness enhancement
fixed-Δ_0 control	weak/sign-sensitive control	universal enhancement
pairing family	tested orbital-pairing family	all possible pairing channels
momentum grids	selected sign/scale stability	continuum-limit value
robustness matrix	production-scope $n_{\text{keep}} = 6, 8$ stability	all-truncation independence
filling variable	central flat-band crosswalk	device density calibration
absolute units	benchmark self-audit only	experimental comparison
valley partner	`tr_sewn` diagnostic convention	valley-basis-independent stiffness claim

3. a final absolute stiffness convention only before any direct experimental comparison;

4. a device-level filling calibration beyond the central-flat-band crosswalk reported here.

These limitations do not invalidate the normalized-response mechanism claim, but they define the boundary of the present PRB mechanism submission. Direct experimental-comparison claims require further convergence, calibration, and absolute-convention work.

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DECLARATIONS

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Data availability: All data and code supporting this work are available from the corresponding author upon reasonable request. The manuscript source, generated data tables, configuration files, plotting scripts, validation scripts, and local package manifests are maintained in the MATBG research repository at https://github.com/JackZH26/MATBG. The

CSV files and scripts referenced in the manuscript and Supplemental Material reproduce the reported tables and figures.

Ethics approval: Not applicable.

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